

Automated Irrigation System Using Image Scanning

Abstract— Agriculture is a crucial sector that requires efficient management of resources such as water and pesticides to ensure optimal crop health and yield. This paper presents a smart automated irrigation and spraying system that integrates image scanning technology to optimize water and pesticide application. A high-resolution camera captures images of plant leaves, which are processed using an image recognition application to determine plant health and detect diseases or deficiencies [1][2]. In addition, humidity and soil moisture sensors provide real-time environmental data, which, in conjunction with image analysis, informs a control system to regulate water and pesticide distribution [3][4]. This approach minimizes resource wastage, reduces labor costs, and enhances sustainable farming practices.

Keywords— *Smart irrigation, IOT, image processing, crop health monitoring, automated spraying, precision agriculture*

I. INTRODUCTION

Water management and pest control are vital for sustaining crop yield and ensuring food security. Traditional irrigation and pesticide application methods often result in inefficiencies, leading to resource overuse or underuse. This project introduces an intelligent irrigation and spraying system that utilizes image processing technology to determine plant needs and automate resource delivery [5][6]. The system ensures timely and precise application of water and pesticides, thereby enhancing plant growth and reducing environmental impact.

II. METHODOLOGY

A. Image Capture and Processing

The system uses an image processing application to analyse these images, which enables detection of plant health status, disease symptoms, and nutrient deficiencies. The image processing algorithm extracts features like colour patterns, leaf textures, and geometrical shapes, which are then compared with a pre-trained dataset to identify potential issues such as fungal infections, pest damage, or water stress [10].

For plant disease detection, a convolutional neural network (CNN) is employed. The CNN model is trained on a dataset of labelled plant diseases, and once trained, it classifies the plant status. Additionally, the system utilizes algorithms to assess the leaf's deficiencies[11].

B. Humidity and Soil Moisture Sensing

A set of sensors measures environmental conditions like humidity and soil moisture. The humidity sensor monitors the moisture level in the air, while the soil moisture sensor detects the water content in the soil [12]. The data from these sensors are transmitted to a microcontroller, such as an Arduino, for real-time processing. The controller then makes decisions based on predefined thresholds to trigger irrigation or spraying mechanisms [8].

C. Water and Pesticide Supply Control

The system automates water and pesticide distribution through the control system. Once the image processing and sensor data indicate the need for irrigation or pesticide

application, the system activates a pump connected to a water reservoir or a pesticide tank. The microcontroller regulates the flow, ensuring the correct amount of water or pesticide is applied to the affected plants [13].

The system uses a proportional-integral-derivative (PID) controller to modulate the flow rate of the pump, ensuring optimal application without overuse. Additionally, actuators are placed near the crops for efficient spraying in precise locations, reducing both chemical usage and water wastage [14]

D. Design and Framework of the Image Processing Web Application

The proposed automated irrigation and spraying system consists of three main components: the Image Processing Module, the Sensor Network, and the Automated Water and Pesticide Control System. These components work in an integrated manner to ensure efficient resource management and crop health monitoring

1. Gathering and Preprocessing Data for Plant Disease Detection:

The foundation of a plant disease detection system lies in collecting an extensive dataset of around 30,000–40,000 images of various plants and their diseases. This dataset is enriched by including original, augmented, and cleaned images. Augmentation techniques such as rotation and flipping create additional variations to improve model robustness. Cleaning ensures the removal of noise, enhancing the clarity of images for more accurate analysis. The quality and diversity of this dataset are critical for training a high-performing machine learning model.

2. Organizing and Sorting Images by Plant and Disease Categories:

After collecting the images, they are systematically sorted into categories based on the plant species and disease type. This classification allows each plant-disease pair to be well-represented within the dataset. Organizing the images into classes is vital for effective training, enabling the machine learning model to learn from a wide range of examples. The sorted images are stored in distinct folders or datasets, ensuring easy access during the training phase, thereby improving the model's ability to classify different plant diseases accurately.

3. Uploading Data into Machine Learning Models: Plant and Disease Classification

The sorted images are then uploaded into machine learning models designed to classify plant species and their associated diseases. Each plant-disease combination is assigned to a specific class within the model. The data is split into training and testing sets to allow the model to learn effectively while

maintaining accuracy. This organized approach helps streamline the classification process, ensuring that the model can identify patterns in various plant species and diseases.

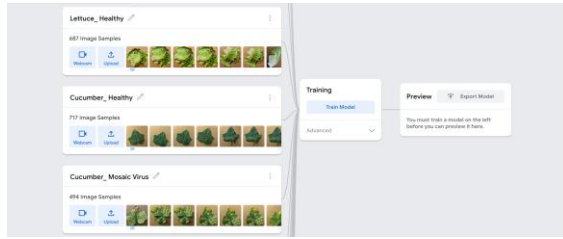


Fig. 1. Uploaded Datasets for ML model

4. Training the Machine Learning Models for Plant Disease Detection

Once the data is ready, the machine learning models undergo training to detect patterns related to different plant diseases. Convolutional Neural Networks (CNNs) are commonly used in this phase, as they excel at processing image data and extracting relevant features. Hyperparameters such as learning rates and batch sizes are fine-tuned throughout the training to optimize model performance. The goal is to build a model that can accurately predict plant diseases across various species with high precision.

5. Testing the Performance of Trained Machine Learning Models

The trained models are subjected to testing using unseen images to evaluate their performance. Metrics like accuracy, precision, and recall are calculated to gauge how well the models perform in identifying plant diseases. This phase helps reveal any areas where the models may need further refinement. By addressing these weaknesses, the models become more reliable for real-world deployment, ensuring consistent and accurate disease detection

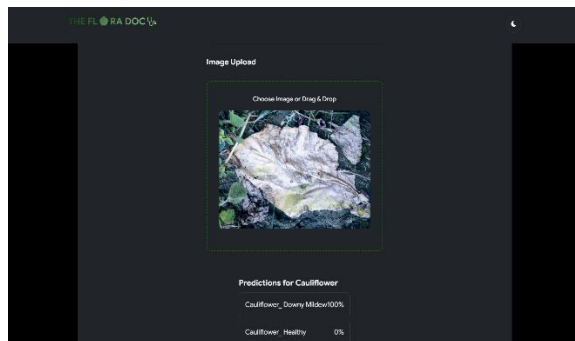


Fig. 2. Testing Accuracy of the Model.

6. Developing a Web Application for Plant Disease Detection Using HTML, CSS, and JavaScript

A web application is created using HTML, CSS, and JavaScript to provide users with a simple interface for interacting with the machine learning models. Users can upload plant images from their devices, and the app processes the input to detect the disease and plant species. The interface is designed to be intuitive and responsive, allowing for easy use on various devices. The app's primary focus is on delivering a smooth user experience while offering clear visual feedback.

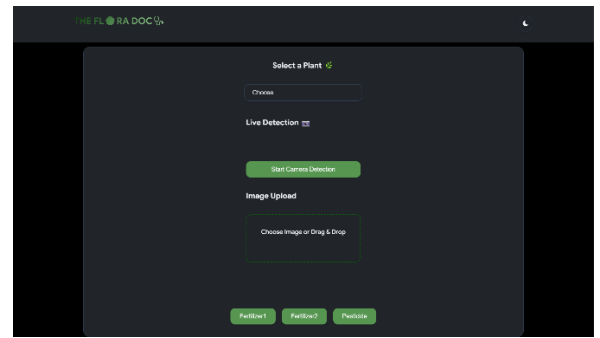


Fig. 3. The Disease Detection Webpage

7. Integrating Machine Learning Models with a Web-Based Interface

The final step involves integrating the trained machine learning models into the web application. This integration allows users to receive real-time predictions by uploading images of plants. JavaScript facilitates communication between the front-end and back-end, enabling seamless data flow between the user interface and the machine learning models. The result is a user-friendly tool that provides immediate and accurate plant disease diagnoses, making advanced agricultural diagnostics accessible to a broader audience

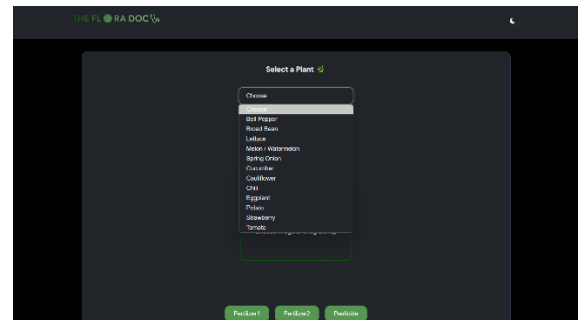


Fig. 4. Several Different ML Models Integrated into the webpage

8. Future Scope of AI-Based Plant Disease Detection Systems

The future of AI-based plant disease detection includes expanding datasets to cover more species and diseases, improving model accuracy and generalization. Cloud deployment will increase accessibility and scalability [15]. Integration with automated irrigation and pesticide systems can streamline farming practices, optimizing resource use. Advances in machine learning, like deep learning, promise faster, more accurate detection, while IoT integration could enable real-time crop monitoring, enhancing overall farm management and sustainability [16].

E. System Architecture of Mechanical Parts.

The mechanical design of the proposed automated irrigation and spraying system plays a crucial role in ensuring efficient operation, durability, and adaptability to various agricultural environments. The system consists of a structural framework, a water and pesticide distribution

mechanism. These mechanical components ensure precision, reduced wastage, and sustainability in management of resources.

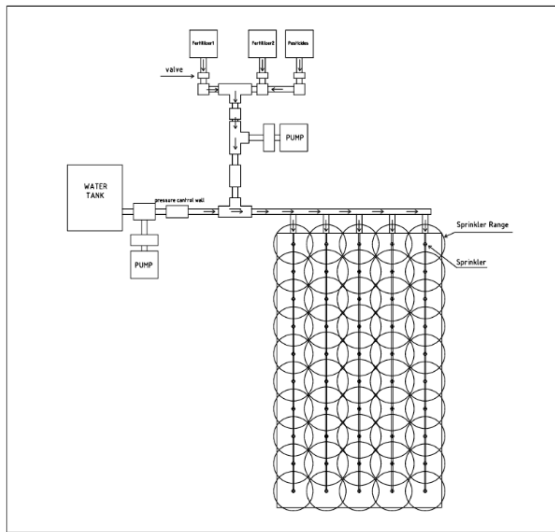


Fig. 5. Diagram of System Layout

1. Mechanical Design of Sprayer System

The sprayer mechanism is a crucial component that ensures precise pesticide application based on image analysis and sensor data. The following mechanical factors were considered in designing the sprayer:

- **Type of Sprayer:** The system incorporates a high-pressure sprayer to ensure even distribution of pesticides on plant surfaces.
- **Nozzle Selection:** The sprayer system is equipped with flat-fan nozzles for uniform coverage and nozzles for targeted spraying. These nozzles ensure fine reducing pesticide overuse and drift.

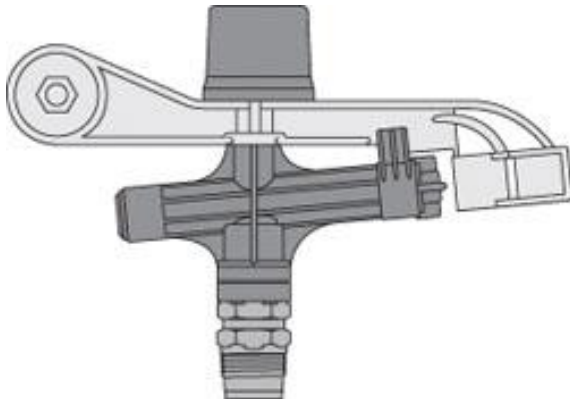


Fig. 6 (a). Sprinkler Nozzle

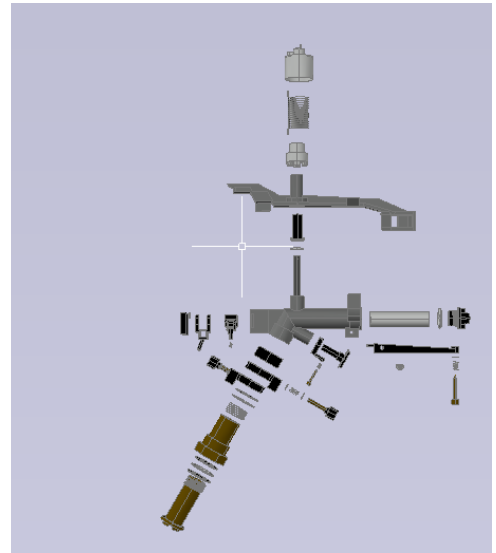


Fig.6 (b). Sprinkler Nozzle

- **Mounting Mechanism:** The spraying arm is adjustable to target plants at different heights, making it adaptable to multiple crop varieties.
- The sprayer is connected to a pesticide reservoir through a Centrifugal Pump and pressure control wall ensuring controlled flow rates. The pressure regulation system maintains a consistent 2–3 bar pressure for uniform spraying.

2. Fluid Flow and Pressure Regulation

To optimize irrigation and spraying efficiency, the system includes a fluid distribution unit that regulates pressure, flow rate, and liquid droplet size. Key aspects include:

- **Pump Selection:** The system integrates a centrifugal water pump (for irrigation and pesticide) to ensure proper distribution.



Fig. 7 (a). Centrifugal Pump

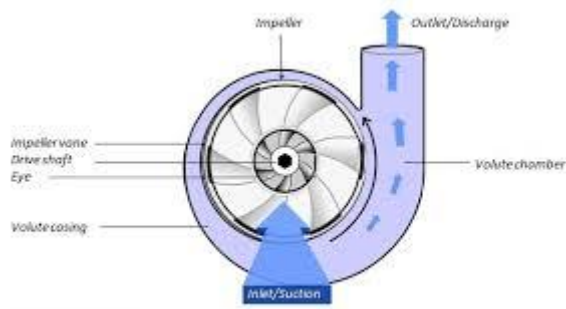


Figure 2. Volute case design

Fig. 7 (b). Parts of a Centrifugal Pump

- Pressure Regulation:
Pressure control wall will control pressure in pumps.
Insuring constant pressure throughout the line.

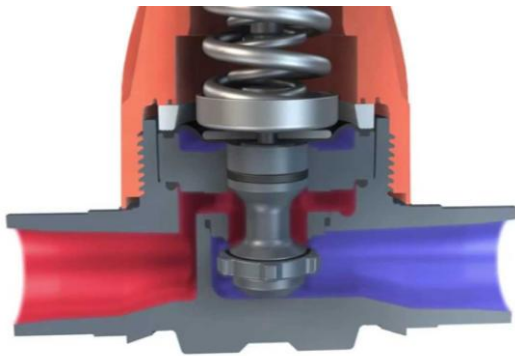


Fig. 8. Pressure Control Valve

3. Wetting Patterns in Sprinklers

The wetting pattern from a single rotary sprinkler is not very uniform. For good uniformity, several sprinklers must be operated close together to overlap their patterns. The overlap should be at least 65% of the wetted diameter for good uniformity. This determines the maximum spacing between sprinklers.

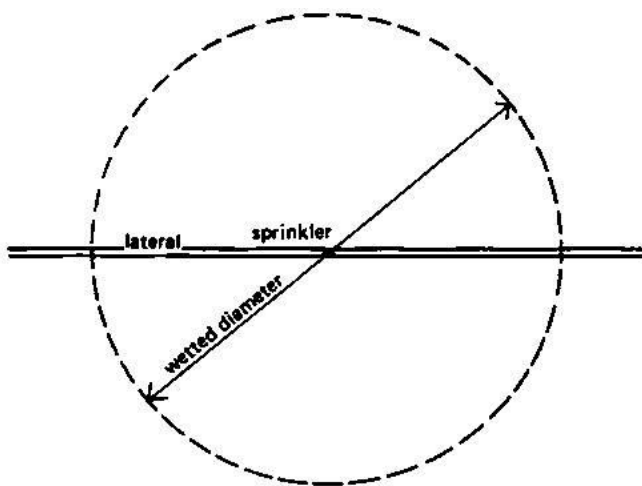


Fig. 8. Wetting pattern for a single sprinkler (TOP VIEW)

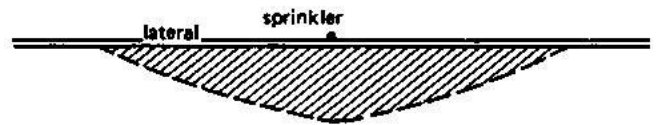


Fig.9. Wetting pattern for a single sprinkler (SIDE VIEW)

F. System Architecture for Electrical Connections

1. Detailed Electric Components and Circuit Connections:

Microcontroller (Arduino):

This will act as the central control unit for the entire system. It will receive data from sensors and image processing modules and send commands to various actuators (valves, pumps, motors). Arduino Pins will be connected to relay modules or transistor switches to control high-power components.

Water, Fertilizer, and Pesticide Tanks (4-Total):

Each tank will have a solenoid valve (normally closed) connected to a pump motor or water source. The solenoid valve will be controlled by the microcontroller to allow the liquid to flow when required. Pump Motors (DC or AC, depending on the system design) will be used to pump liquids from the tanks to the sprayers. Flow sensors could be installed to measure the flow of liquids.

Sprinklers/Sprayers:

Depending on the tank (water, fertilizer, or pesticide), the system will control a series of sprayers connected to solenoid valves. Solenoid Valves control the release of the liquids. The sprayers could be connected to the solenoid valve through a pipe or tube system. Relay modules or motor drivers will be used to control the solenoid valves.

Sensors:

Soil Moisture Sensor: Measures the moisture level of the soil. If moisture is low, the microcontroller will activate the water tank.
Temperature and Humidity Sensors: Helps in understanding the environmental conditions and adjusting watering/fertilizer needs accordingly.
Flow Sensors: Can be used to ensure that the proper amount of liquid is flowing during watering, fertilization, or pesticide application.

Relay/Transistor Driver Circuits:

Relay Modules or Transistor Drivers will be used to control high-power components like pumps and solenoid valves. These components are necessary because the microcontroller can't

directly control high-power devices like motors and valves.

Power Supply:

Power Supply for the microcontroller, sensors, valves, pumps, and motors. For instance, the pumps, motors, and solenoid valves would need a separate power supply (like 12V or 24V, depending on the specifications of the components).

G. Working Mechanism

1. Soil Moisture Monitoring & Watering Process

Step 1: Collecting Soil Moisture Data

- The Soil Moisture Sensor is inserted into the soil to measure moisture levels.
- The sensor sends an analogue signal to the Arduino, which converts it to a digital value (0–1023).
- The Arduino compares this value with a predefined threshold level (e.g., if moisture is below 40%, watering is needed).

Step 2: Activating the Watering System

If soil moisture is below the threshold, the Arduino performs the following:

- Activates the relay module that controls the solenoid valve of the water tank.
- Turns ON the pump motor, allowing water to flow through the pipe system to the sprinklers.
- The flow sensor ensures that the correct amount of water is dispensed.
- Water is sprayed through connected sprinklers/sprayers onto the field.

Step 3: Stopping Watering Process

- The system continuously monitors soil moisture levels.
- Once moisture reaches the required level, the Arduino:
- Turns OFF the pump motor.
- Closes the solenoid valve, stopping the water flow.
- This prevents overwatering and conserves water.

2. Fertilizer and Pesticide Spraying

I. Manual Button Activation: The spraying of fertilizers or pesticides is initiated by manual button presses. There are three separate buttons, each corresponding to:

- Fertilizer 1 button
- Fertilizer 2 button
- Pesticide button

When any of these buttons are pressed, the corresponding solenoid valve opens, allowing the fertilizer or pesticide to flow into the pipe.

ii. Controlled Dispensing: The system is designed to dispense the fertilizers or pesticides in a controlled

manner, ensuring a precise ratio of the chemical to water. This is achieved by,

- Flow sensors are installed in the outlet pipes of the fertilizer/pesticide tanks to measure the volume of liquid passing through. The flow sensors monitor the amount of liquid being dispensed, ensuring that the desired volume (e.g., 200ml of fertilizer/pesticide) is released.
- Water is simultaneously pumped into the pipe, and its flow is controlled by a separate solenoid valve and flow sensor to maintain the proper ratio (e.g., 1000ml of water).

iii. Mixing of Fertilizer/Pesticide with Water: After the respective button is pressed:

- The solenoid valve of the selected fertilizer/pesticide tank opens, allowing the liquid to flow into the pipe.
- At the same time, the water pump activates, and water is added to the pipe through a separate pipe connected to the same sprinkler system.
- The ratio (e.g., 200ml of pesticide/fertilizer per 1000ml of water) is controlled using solenoid valves and flow sensors for both water and fertilizer/pesticide lines.
- The liquids mix within the pipe as they flow toward the sprinkler nozzle.

iv. Spraying Process: The mixed solution of water and fertilizer/pesticide is then sprayed through a sprinkler nozzle attached to the pipe. The pressure in the pipe is controlled by a pressure regulator to ensure an even and consistent spray pattern. The spraying continues until the required volume of the fertilizer/pesticide has been dispensed, as measured by the flow sensors.

v. Automatic Shut-off: Once the system has dispensed the predetermined amount of fertilizer or pesticide (e.g., 200ml), the solenoid valve for the selected tank closes, stopping the flow of liquid. The pump motor for the fertilizer/pesticide also stops. The system can then return to a standby mode, awaiting further button presses or automated commands for water spraying.

vi. Ratio Control: The system ensures that the ratio of fertilizer/pesticide to water is precise. The flow sensors in both the fertilizer/pesticide and water lines continuously measure the volume of liquid being dispensed. If the ratio deviates from the desired value, the system can adjust the solenoid valves and pump speeds to correct it. This ensures the right concentration of chemicals is applied to the crops.

3. Summary of Working Flow

The system starts by initializing sensors, pumps, valves, and the microcontroller. It then checks soil moisture levels, and if they are low, water is pumped from the tank to the sprinklers.

Watering continues until the soil reaches the required moisture level, after which the system stops. When fertilization or pesticide spraying is needed, either based on a predefined schedule or sensor data, the system activates the appropriate tank. The liquid is sprayed using the pump and solenoid valve system, with flow sensors ensuring accurate distribution.

III. RESEARCH

After disease detection in plants, the following section deals with selection of water-soluble pesticides for disease control in crops, e.g., potatoes, tomatoes, and eggplants. It gives a list of pesticides, their type, and target disease, along with a consideration of irrigation parameters like pressure and flow rate for effective distribution of nutrients. The section ends with the identification of individual diseases to be treated by the automated system.

A. Water-Soluble Pesticides

TABLE I.

Pesticide/Fungicide Name	Type	Crops	Target Pests/Diseases
Imidan 70-W (Phosmet)	Insecticide	Potato, tomato, pepper, eggplants	Aphids, beetles, caterpillars
Phosphorous Acid Fungicides (ProPhyt, Rampart)	Fungicide	Pepper, Eggplant, Cucumbers	Phytophthora, blight, downy mildew
Xylar FC (Metalaxyl)	Fungicide	Potato, Cucumbers	Pink rot, Pythium leak
Elatus (Benzovindiflupyr + Azoxystrobin)	Fungicide	Potato	Rhizoctonia, other soil-borne diseases
Biopesticides (GreenFurrow BacStop)	Fungicide	Potato	Common scab, early blight, late blight

B. Pressure and Flow Rate According to Different Plants.

TABLE II.

Crop	Pressure (Bars)	Flowrate (Liters per Hour - LPH)
Black Gram	2.8	500 LPH
Bell Pepper	1.4 - 2.8	200 LPH
Cucumber	0.3 - 1.7	95 - 375 LPH
Eggplant	1.4 - 2.8	300 - 500 LPH
Strawberry	5.5	400 - 800 LPH
Potato	2 - 2.5	100 - 200 LPH
Melon	1.5 - 2.5	250 - 400 LPH

Watermelon	1.5 - 2.8	300 - 500 LPH
Chili	1.4 - 2.2	200 - 400 LPH
Spring Onion	0.6 - 1.5	75 - 150 LPH

C. Disease That can be Identified Using This System

This system can effectively cure several plant diseases across different crops. For black gram, it addresses leaf crinkle. In cauliflower, it treats diseases such as black rot, downy mildew, and bacterial spot. For chili, the system can combat leaf curl, leaf spot, leaf whitefly, and leaf yellowing. In eggplant, it targets insect-based diseases, leaf spot, mosaic virus, white Mold, and wilted leaf. Additionally, for potatoes, it treats early blight and late blight, while in strawberries, it helps control leaf scorch.

IV. CONCLUSION

In summary, the new smart automated spraying and irrigation system presents a breakthrough innovation in contemporary agriculture through the use of image scanning technology and real-time sensor information in precise water and pesticide application. The system improves crop health monitoring, maximizes the use of resources, and ensures sustainable agriculture through water and chemical conservation. Automation of the processes reduces labour expenditure and facilitates timely intervention against plant disease and deficiency [7]. The system's scalability and flexibility across crops make it an ideal solution to the future challenge of sustainable agriculture..

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REFERENCES

- [1] K. S. P. Reddy and P. S. Babu, "A novel improved irrigation system using image processing and wireless technology," 2016.
- [2] H. A. M. Tran, H. Q. T. Ngo, T. P. Nguyen, and H. Nguyen, "Design of green agriculture system using internet of things and image processing techniques," in Proc. 4th Int. Conf. Green Technology Sustainable Development (GTSD), 2018, pp. 28–32.
- [3] S. Kumawat, M. Bhamare, A. Nagare, and A. Kapadnis, "Sensor based automatic irrigation system and soil pH detection using image processing," Int. Res. J. Eng. Technol., vol. 4, pp. 3673–3675, 2017.
- [4] M. Polic, M. Car, J. Tabak, and M. Orsag, "Robotic irrigation water management: Estimating soil moisture content by feel and appearance," in Proc. 19th Int. Conf. Ubiquitous Robots (UR), 2022, pp. 16–22.
- [5] U. Ahmad, D. M. Subrata, and C. Arif, "Speaking plant approach for automatic fertigation system in greenhouse," arXiv preprint arXiv:1303.1869, 2013.
- [6] N. A. M. Leh, M. S. A. M. Kamaldin, Z. Muhammad, and N. A. Kamarzaman, "Smart irrigation system using internet of things," in

Proc. IEEE 9th Int. Conf. System Eng. Technol. (ICSET), 2019, pp. 96–101.

- [7] P. Suganya, A. Subramanian, B. Yagneshwaran, F. Khan, and S. R. Sushmitha, “Automatic irrigation system using IoT,” *Int. Res. J. Eng. Technol.*, vol. 4, no. 10, 2025.
- [8] D. Maheswari, R. Harish, S. Sowmiya, and M. K. Aiswarya, “Smart crop protection & automatic irrigation system using IoT & ML.
- [9] J. J. Amalraj, S. Banumathi, and J. J. John, “A study on smart irrigation systems for agriculture using IoT,” *Int. J. Sci. Technol. Res.*, Dec. 2019.
- [10] F. Behzadipour, M. G. N. Raeini, S. Abdanan Mehdizadeh, M. Taki, B. Khalil Moghadam, M. R. Z. Bavani, and J. Lloret, “A smart IoT-based irrigation system design using AI and prediction model,” *Neural Comput. Appl.*, vol. 35, no. 35, pp. 24843–24857, 2023.
- [11] B. Nsoh, A. Kamtbo, H. Guo, D. M. Heeren, H. N. Nakabuye, X. Qiao, Y. Ge, D. R. Rudnick, J. Wanyama, E. Bwambale, and S. Kiraga, “Internet of Things-based automated solutions utilizing machine learning for smart irrigation: A systematic review,” *Sensors*, vol. 24, no. 23, p. 7480, 2024.
- [12] B. R. Puneeth, A. P. Supravi, P. S. Netravathi, and S. R. Devadiga, “IoT-based smart irrigation management system: Design and implementation,” *Int. Res. J. Modern. Eng. Technol. Sci.*, May 2023.
- [13] M. S. Vijaya, S. Balakrishna, T. S. Rekha, P. A. Madhulatha, Y. N. Joshnavi, and P. Dedeepya, “Automated irrigation management system using IoT,” *J. Emerg. Technol. Innov. Res.*
- [14] V. R. Balaji, “Smart irrigation system using IoT and image processing,” *Int. J. Eng. Adv. Technol.*, vol. 8, no. 6S.
- [15] D. M. Kharat and P. M. Jawandhiya, “Intelligent IoT-based automated irrigation system: A review,” *Int. J. Res. Advent Technol., Convergence 2018 Conf. Proc.*
- [16] K. Balakumaran, “IoT-based automated irrigation system for optimal water use in agriculture,” *Int. J. Novel Res. Dev.*, vol. 8, no. 8